Biology - Undergraduate Level Assessment 6 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. What hormone is responsible for metamorphosis in amphibians?
   (a) growth hormone
   (b) estrogen
   (c) thyroxine
   (d) insulin
   (e) prolactin

2. The ability of the brain to detect differences in stimulus intensity is best explained by the fact that which of the following varies with the stimulus intensity?
   (a) The number of synapses crossed
   (b) The speed of the action potential
   (c) The amplitude of the action potential
   (d) The number of neurons activated
   (e) The refractory period of the neuron

3. Two individuals, one with type B blood and one with type AB blood, have a child. The probability that the child has type O blood is
   (a) 50%
   (b) 100%
   (c) 0%
   (d) 10%
   (e) 75%

4. Keystone species are thought to have profound effects on the structure and composition of ecological communities because they
   (a) are more abundant than most other species in their communities
   (b) tend to reduce diversity by eliminating food resources for other species
   (c) are the largest species in their communities
   (d) are the most aggressive species in their communities
   (e) are the only species capable of reproduction

5. A deficiency in which endocrine gland would result in tetany?
   (a) Pancreas

(b) Thymus
(c) pituitary gland
(d) adrenal gland
(e) thyroid gland

True / False (10 marks)

Answer True or False

6. True or False: All tissue types in higher plants have identical functional roles, making classification straightforward.
7. True or False: Organ transplants generally fail due to the lack of skilled surgical expertise during the transplant procedure.
8. True or False: The pH of both lakes will decrease in response to acid rain, indicating increased acidity in both environments.
9. True or False: A deletion of a nucleotide triplet after the start codon typically results in the most severe disruption of the protein sequence.
10. True or False: Mules have a relative evolutionary fitness of zero due to their inability to produce viable offspring.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how the removal of product C in the enzyme-catalyzed reaction $A + B \rightarrow C$ can shift the equilibrium to favor increased production of C.
12. Discuss how two minks with brown fur can produce offspring with silver fur, including the genotypes involved and the principles of inheritance that apply to this scenario.

End of Paper